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ARTICLES

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Scalable ultrasmall three-dimensional nanowire transistor probes for intracellular recording

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Introduction

What is the problem this paper is trying to solve?

To build scalable tools that enable high spatiotemporal resolution intracellular potential recording

Is this problem important?

Intracellular probes are required towards deeper understanding of electrogenic cells and their networks

What have others done to solve this?

1) Patch-clamp electrodes
2) Scalable on-chip micro/nano-structured electrode arrays
3) 3D nanowire field-effect transistor (FET) probes

→ Rely on one-by-one fabrication that is difficult to scale up

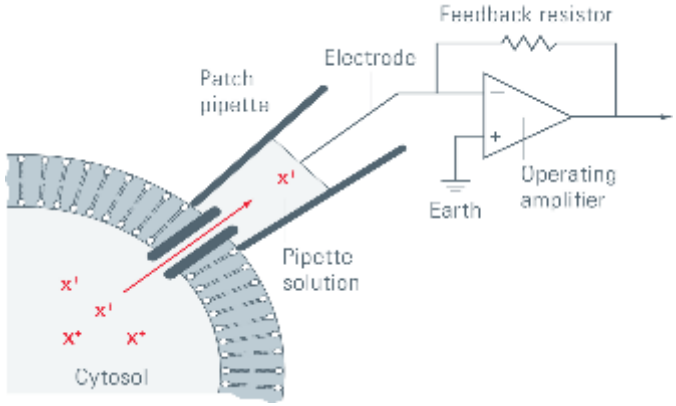
Background Information

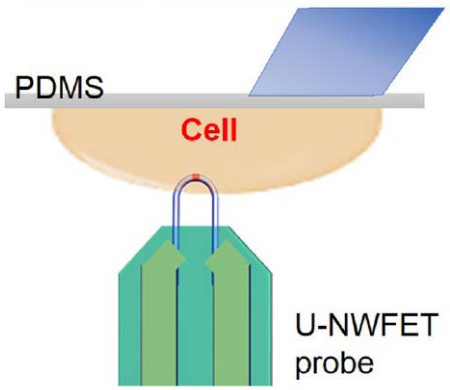
Patch-clamp electrode

- Used to study ionic currents in individual isolated living cells, tissue sections, or patches of cell membrane
- The current passing across the membrane is controlled by the experimenter and the resulting changes in voltage are recorded

Nano FET probe

- Biomimetic phospholipid modification to provide high-resistance membrane seals and
- Achieve direct access to the cell interior





Introduction

What has this research done differently to solve this?

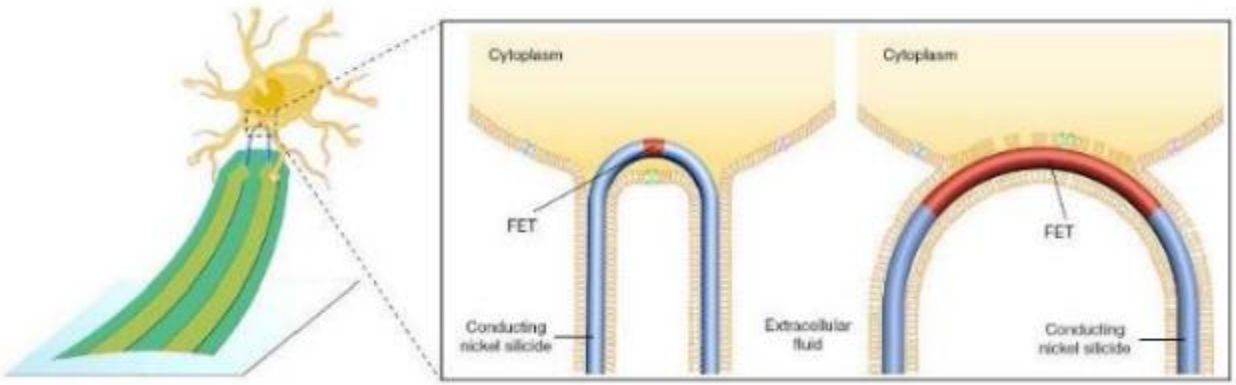
1) Large-scale and shape-controlled deterministic assembly

2) Spatially defined solid-state transformation

Results

1) Scalable nanowire FET probe arrays with controllable tip geometry and sensor size enabling high amplitude intracellular action potentials recording

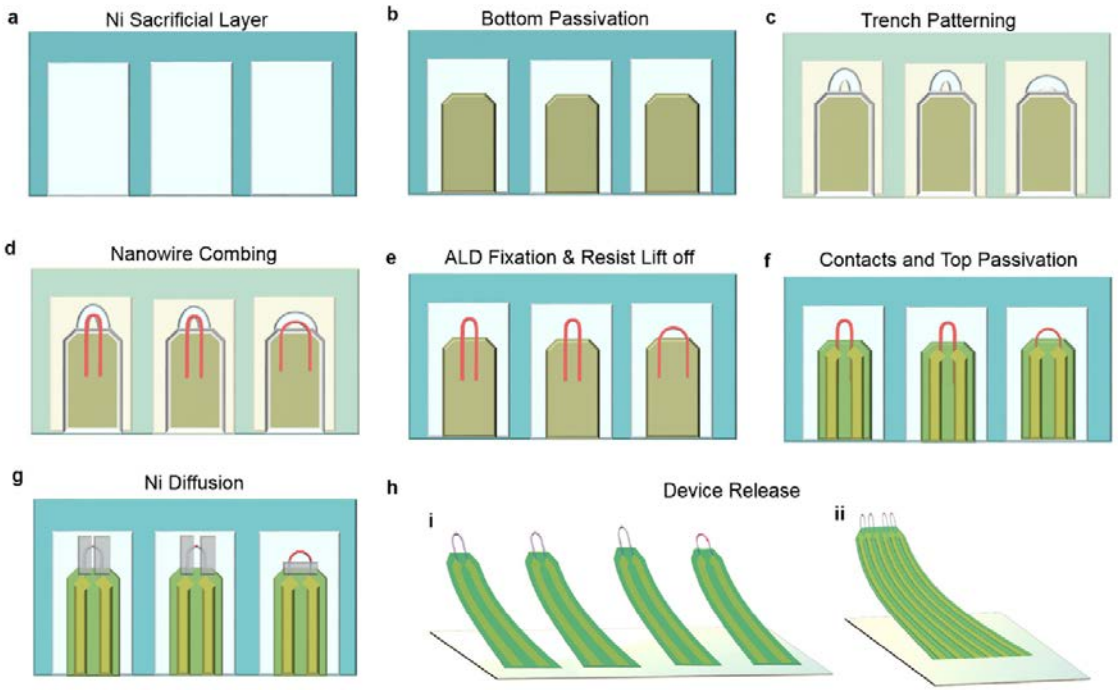
2) Investigation on how the size and geometry of nano-probes influence intracellular recording



Fabrication

Fabrication process

1) Ni sacrificial layer pat. and dep.
2) Si_3N_4 passivation layer pat. and dep.
3) Pat. on PR
4) Shape-controlled deterministic NW assembly
5) Dep. of Al_2O_3
6) Interconnects pat. and dep.
7) Ni diffusion layer pat. and dep.
8) Ni sacrificial layer and diffusion layer removal

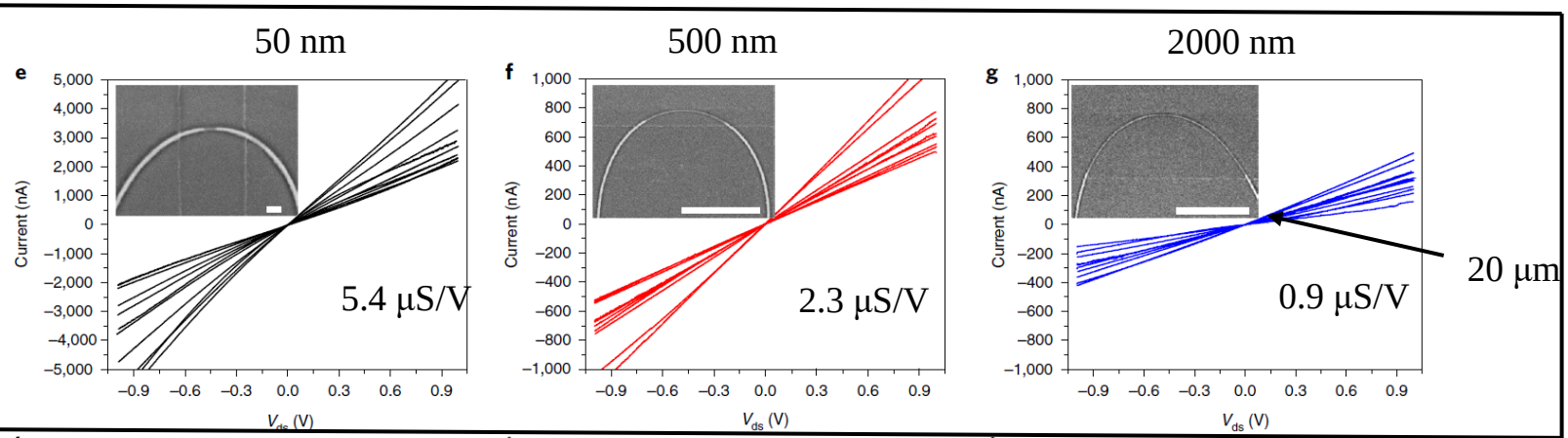


Characterization

Sensor properties

1) Current vs. drain-source voltage – dry state
2) Conductance vs. gate voltage – aqueous state
3) Transconductance vs. ROC
4) Detection sensitivity: transconductance and measured noise values → allow 1-10mV detection

1)



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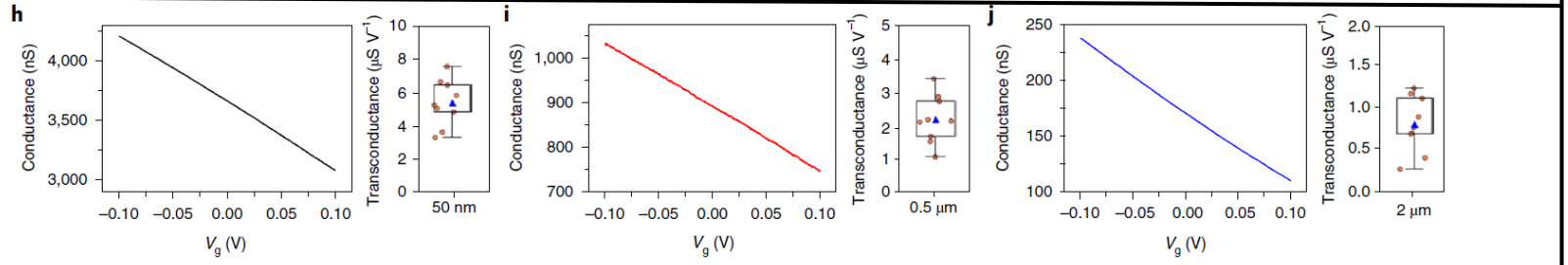
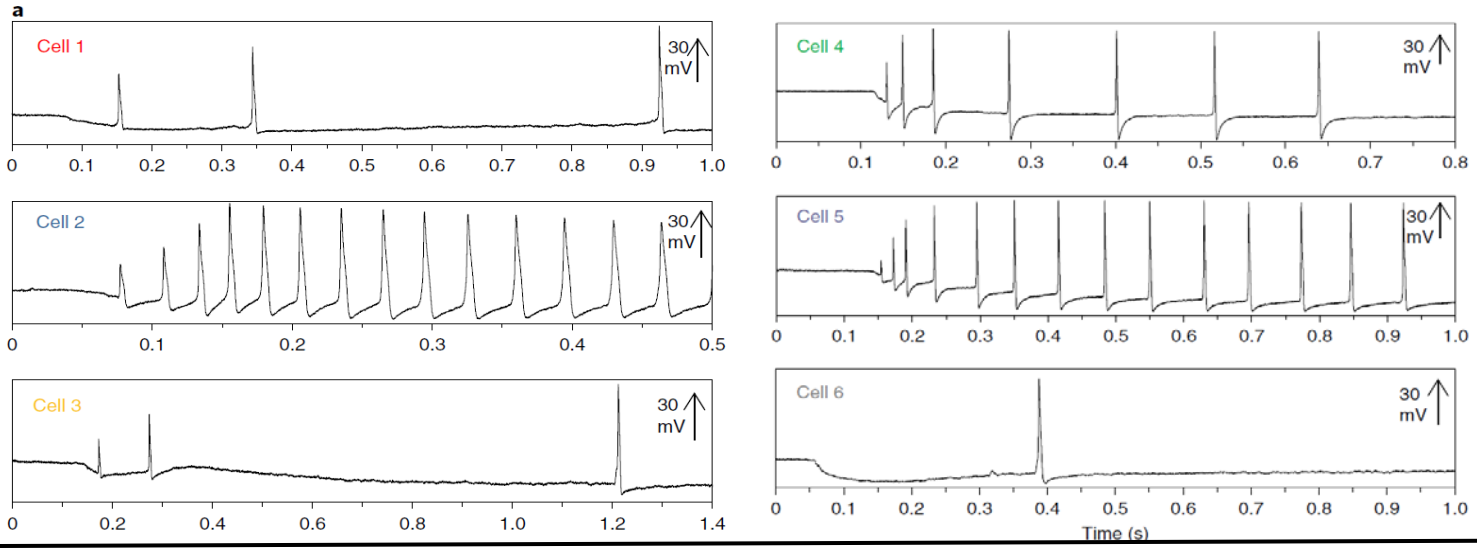
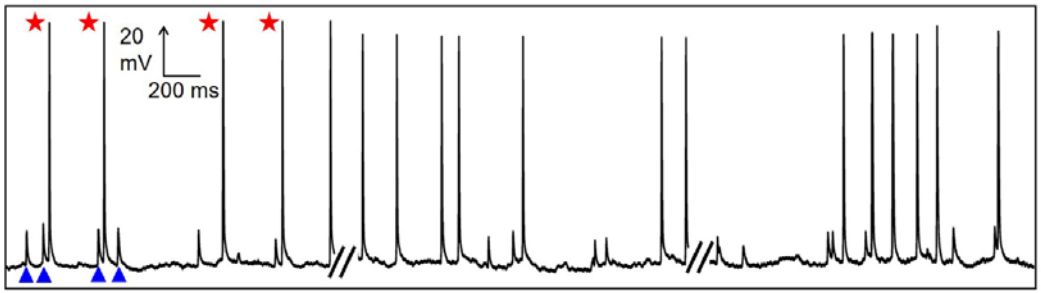


→ consistent

Intracellular Recording Validation

1. Full amplitude intracellular action potential recording

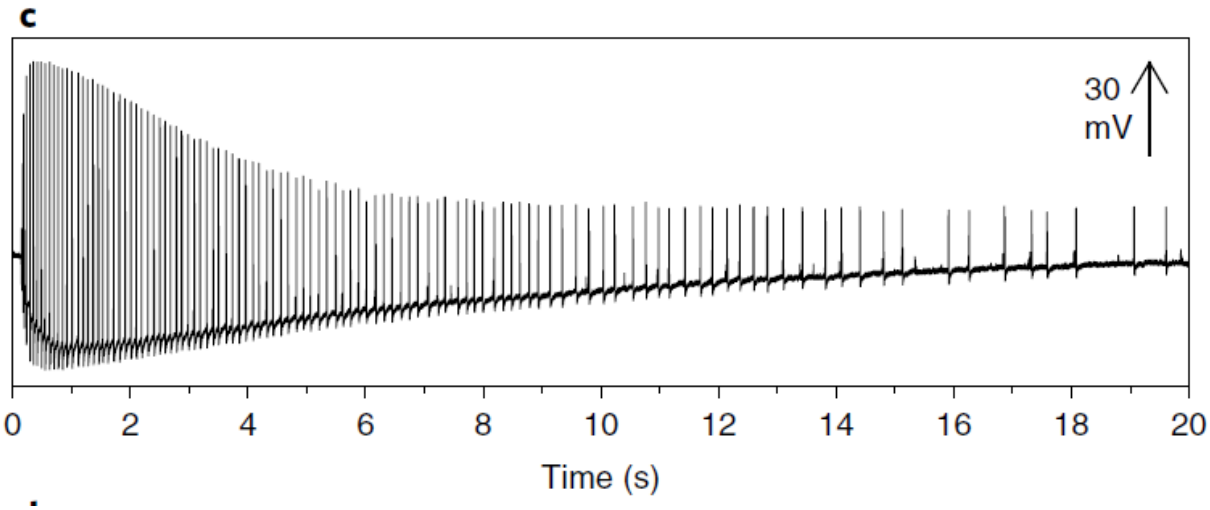
- U-NWFET - 50nm length, 0.75 μm ROC for measuring six DRG neurons
- ➢ 60-100 mV peaks, SNR = 115

Exp	
Patch clamp	

Intracellular Recording Validation

1. Full amplitude intracellular action potential recording

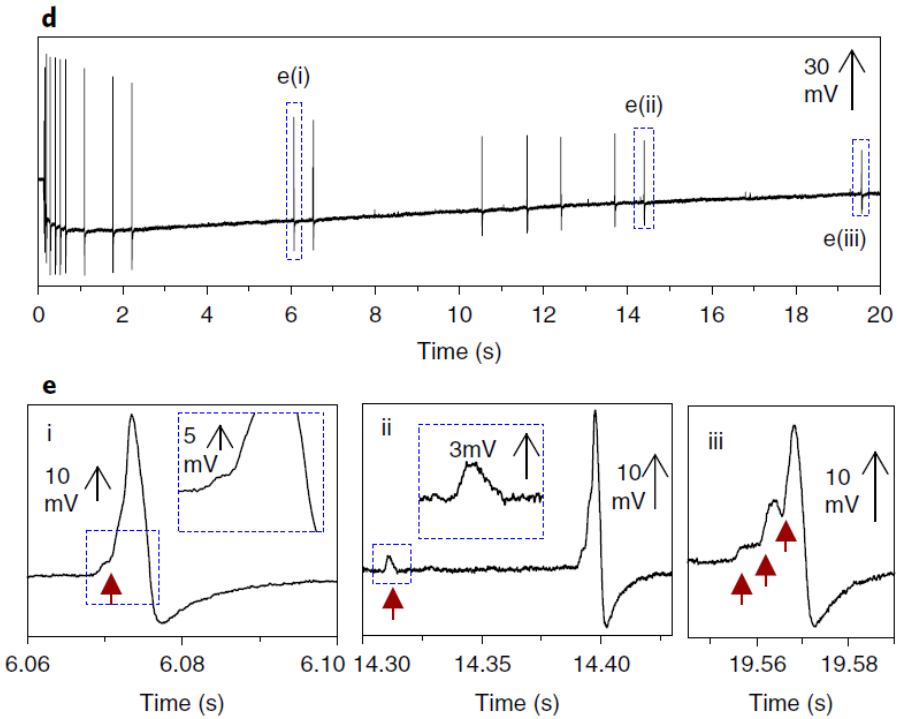
- Limitations
  - Shift in baseline
  - Decrease in recorded AP amplitude at later times



Intracellular Recording Validation

2. Subthreshold activity observation

- Subthreshold features
  - 5 mV peak immediately before an AP
  - 3 mV peak not associate with an AP

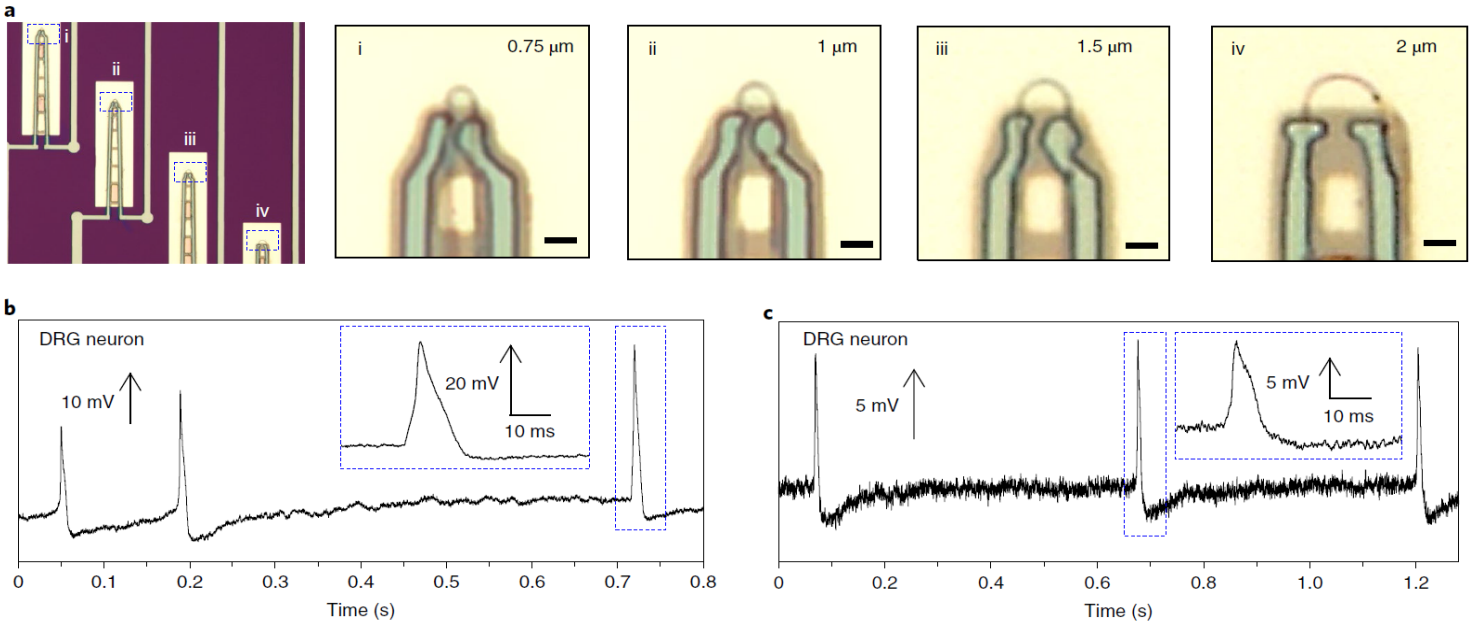


Study of Size and Geometry

1. Geometry effects

- ROC variation: from 0.75 to 2 µm
- Fixed sensor size: 50 nm

➢ Increasing ROC correlates with lower recorded maximum amplitude



ROC: 1 µm

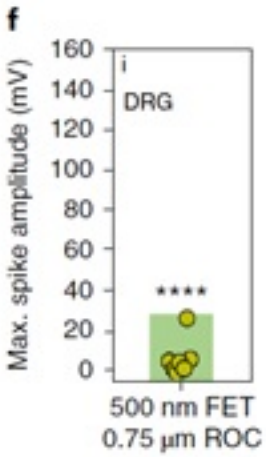
ROC: 1.5 µm

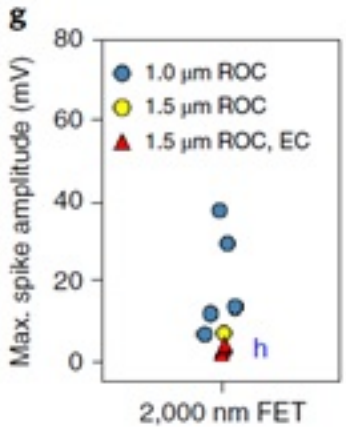
Study of Size and Geometry

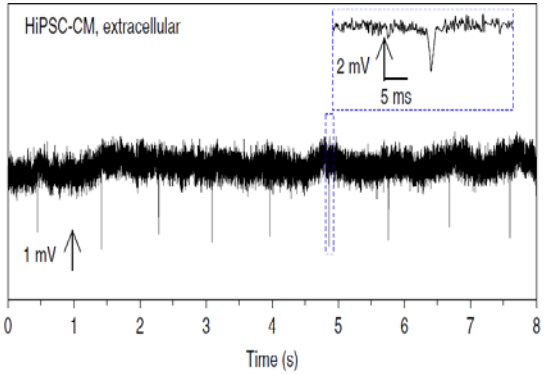
2. Sensor size effects

- Case 1: 500 nm channel length, 0.75 µm ROC
- Case 2: fixed 2000 nm channel length, ROC variation

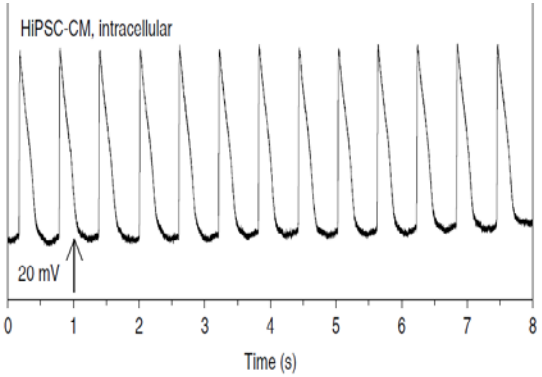
➢ Reducing FET channel length has correlation with increases in maximum measured AP amplitudes







FET: 2000 nm
ROC: 1.5 µm



FET: 50 nm
ROC: 0.75 µm

Study of Size and Geometry

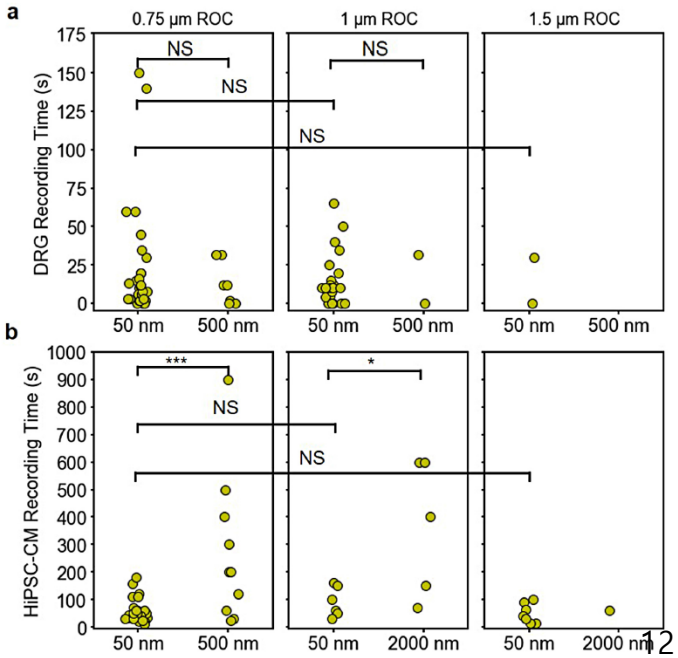
3. Recording duration, ROC, and chan. length relationship

- Variations of ROC was not related to recording duration

➢ Loss of intracellular access is related to curvature-independent factors

- DRG neurons: Channel length was not related to recording duration
- HiPSC-CMs: Increasing channel length attributed in increasing recording duration

➢ Higher probability of maintaining a partially internalized configuration during cell contraction-induced instabilities



Conclusions

Summary

1) Scalable nanowire FET probe arrays with controllable tip geometry and sensor size enabling high amplitude intracellular action potentials recording using
2) shape-controlled deterministic assembly and spatially defined solid-state transformation
3) Investigation on how the size and geometry of nano-probes influence intracellular recording

Challenges

1) Limited number of recording channels
2) Long-term stability of the intracellular recording